

Target-Local Soft-Information Fusion for Communication-Constrained Multi-UAV OTFS-ISAC Sensing

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Abstract—Multi-UAV cooperative sensing gains depend not only on which observations are selected, but also on where their soft reports are fused. We propose a target-local fusion architecture for prior-assisted OTFS-ISAC sensing. Each target is assigned to the UAV nearest its predicted position; a communication-error-calibrated, adaptive coarse-to-fine (C2F) selector then screens and replays target-dependent sensing-reporting candidates using the implemented detector. The fusion rule uses predicted beliefs only and preserves the same finite-blocklength model, 640-bit packet, 2048 channel uses, greedy objective, and resource constraints as the previous canonical system. In 1000 paired trials with 15 UAVs and 10 targets, the resulting V1 system attains $P_D = 0.9734$ (95% CI: [0.9701, 0.9764]) and worst-target $P_D = 0.968$ using 11,993 reports, 7.676 kbit, and 12.793 ms of serial reporting. At identical per-trial resources, its paired gains are 0.0067 over exact-marginal greedy and 0.0085 over sensing-SINR selection, with both 95% confidence intervals above zero. A fusion-rule ablation and six operating-boundary sweeps show that the gain arises from target-related fusion planning jointly with two-stage selection, while large belief errors, strict rate thresholds, and sparse deployments remain the principal limitations.

Index Terms—Multi-UAV networks, OTFS-ISAC, target-local fusion, cooperative sensing, finite-blocklength reporting, coarse-to-fine selection.

I. INTRODUCTION

Integrated sensing and communication (ISAC) allows a wireless network to reuse spectrum, hardware, and signal-processing resources for transmission and environmental sensing [1], [2]. UAV-enabled ISAC further exploits mobility and favorable line-of-sight geometry to extend sensing coverage [5], [6]. When multiple UAVs observe the same target from different bistatic geometries, their soft decisions can provide complementary evidence [3].

This cooperation is not free. Each observation is produced by a sensing chain $i \rightarrow q \rightarrow j$ and must then be delivered from receiver j to a fusion UAV. A strong echo may contribute little if its reporting link is unreliable, its delay-Doppler (DD) energy is poorly captured, or its packet cost exceeds its marginal detection value [9]–[11]. Existing node-selection and backhaul-aware studies address parts of this coupling, while

distributed-detection models often abstract away the bistatic geometry and OTFS DD structure [4], [7], [8].

A further architectural choice has received less attention: *where should reports about each target meet?* A target-agnostic rule that maximizes a UAV’s aggregate incoming rate tends to assign every target to the same central receiver. That concentration changes all reporting rates, finite-blocklength success probabilities, and even the feasible candidate sets seen by the inner selector. Thus, improving only the link-ranking formula cannot correct an unfavorable fusion destination.

We study prior-assisted confirmation or re-detection and introduce target-local fusion planning. For every target, the fusion UAV is chosen using the predicted target position rather than the unknown truth. The resulting target-specific reporting graph is passed to an adaptive two-stage C2F selector whose final greedy replay is aligned with the implemented detector. This separation is deliberate: the fusion rule changes the information available to the selector, whereas the fair marginal-gain objective and resource constraints remain fixed.

Our main contributions are as follows.

- We formulate joint target-specific fusion planning and triplet-level soft-report selection for multi-UAV bistatic OTFS-ISAC. The fusion destination depends only on the predicted target belief, preventing truth leakage into the scheduler.
- We combine this outer planning layer with communication-error calibration, an adaptive coarse frontier, waveform-derived DD refinement, and detector-aligned greedy replay. The inner fair utility and communication price are unchanged, which isolates the architectural role of fusion placement.
- We provide a reproducible V1 evaluation: a 1000-trial paired main comparison, a controlled fusion-rule ablation, reporting-structure diagnostics, and sweeps over belief error, rate threshold, deployment area, communication price, blocklength, and fleet size.
- We identify the operating boundary rather than reporting a single favorable point. The results distinguish measured serial latency from a prospective parallel-scheduling

lower bound and distinguish confirmatory MC=1000 evidence from exploratory MC=100 sweeps.

The remainder of the paper presents the system model in Section II, the target-local C2F method in Section III, the V1 evidence in Section IV, and conclusions in Section V.

II. SYSTEM MODEL

We consider a sensing-centric multi-UAV OTFS-ISAC network, as shown in Fig. 1(a). The UAV and target sets are denoted by $\mathcal{M} = \{1, \dots, M\}$ and $\mathcal{Q} = \{1, \dots, Q\}$, respectively. UAV i and target q have position-velocity pairs $(\mathbf{p}_i, \mathbf{v}_i)$ and $(\mathbf{p}_q, \mathbf{v}_q)$. A cooperative candidate is indexed by the target-dependent triplet (i, j, q) , where UAV i transmits an OTFS-ISAC probing signal, target q reflects it, and UAV j receives the bistatic echo. The inter-UAV reporting topology is represented by a directed graph $\mathcal{G} = (\mathcal{M}, \mathcal{E})$, where $(i, j) \in \mathcal{E}$ if $\|\mathbf{p}_i - \mathbf{p}_j\| \leq d_{\max}$ and $i \neq j$.

We explicitly separate the *sensing* topology from the *reporting* topology. The sensing chain is $i \rightarrow q \rightarrow j$: UAV j receives the echo and therefore *produces* the soft statistic. The statistic is then reported over the link $j \rightarrow f_q$ to a target-specific fusion UAV $f_q \in \mathcal{M}$, which is the single node where all soft statistics about target q meet. Accordingly, a sensing-reporting candidate is the quadruple $(i, j, q; f_q)$, and the communication quantities that matter—reliability $\chi_{j f_q}$, rate $R_{j f_q}$ and delay $T_{j f_q}$ —are properties of the reporting leg (j, f_q) alone, with no dependence on the illuminator i . This separation ensures that the fused statistic is physically well defined: with a naive $j \rightarrow i$ reporting rule, three selected links (i_1, j_1, q) , (i_2, j_2, q) , (i_3, j_3, q) would deliver their statistics to three different UAVs while the fusion step implicitly assumed they had already met. We use a geometry-driven soft-information abstraction rather than a full waveform-level OTFS receiver model. The OTFS structure enters the scheduling problem through DD-bin validity, fractional DD energy loss, same-bin collision, and local DD-grid refinement. These DD-domain quantities determine whether a target-dependent triplet provides useful soft information after reporting and fusion.

A. Communication and Reporting Model

The transmit power of UAV i is split into sensing and communication parts, i.e.,

$$P_i^s = \rho P_i, \quad P_i^c = (1 - \rho) P_i. \quad (1)$$

For a directed inter-UAV link (a, b) , the communication-channel power gain is

$$g_{ab}^c = \left(\frac{\lambda}{4\pi} \right)^2 d_{ab}^{-\alpha_c} \zeta_{ab} |h_{ab}^c|^2, \quad d_{ab} = \|\mathbf{p}_a - \mathbf{p}_b\|. \quad (2)$$

Here λ , α_c , ζ_{ab} , and h_{ab}^c denote the carrier wavelength, path-loss exponent, shadowing, and small-scale fading, respectively.

The conference release uses orthogonal inter-UAV reporting. The communication SINR from UAV i to UAV j is therefore

$$\gamma_{ij}^c = \frac{P_i^c g_{ij}^c}{N_0 + \epsilon_s \sum_{k \in \mathcal{M} \setminus \{i, j\}} P_k^s g_{kj}^c + \epsilon_d P_i^s g_{ij}^c}. \quad (3)$$

The coefficients ϵ_s and ϵ_d represent neighbor-induced sensing-to-communication leakage and intra-link leakage, respectively. Other report payloads do not enter (3), because strictly serial reports cannot interfere with one another.

Shared-spectrum coupling with an orthogonal reporting phase. All UAVs continuously radiate their sensing components over the same propagation field. Soft reports are then exchanged in orthogonal slots. Thus, the communication receiver sees sensing-waveform leakage, while the sensing receiver is limited by residual direct paths from the concurrently radiated sensing waveforms:

$$\begin{aligned} I_{ij}^c &= \epsilon_s \sum_{k \in \mathcal{M} \setminus \{i, j\}} P_k^s g_{kj}^c, \\ I_j^s &= \kappa_{\text{self}} P_j + \kappa_{\text{dc}} \sum_{k \in \mathcal{M} \setminus \{j\}} P_k^s g_{kj}^c. \end{aligned} \quad (4)$$

The phase separation removes circular payload interference without removing ISAC coupling: sensing waveforms still leak into the report receiver and their residual direct paths limit sensing. Here κ_{dc} denotes the residual after reconstructing and cancelling the known illuminator waveforms.

With communication bandwidth $B_c = N \Delta f$, the reporting rate is

$$R_{ij} = B_c \log_2(1 + \gamma_{ij}^c), \quad (5)$$

and a reporting link is feasible when $R_{ij} \geq R_{\min}$.

Finite-blocklength reliability. A soft sensing report is a short control packet. The canonical simulation uses the normal approximation

$$\epsilon_{ij} \approx \mathcal{Q} \left(\frac{C(\gamma_{ij}^c) - \frac{k}{n}}{\sqrt{V(\gamma_{ij}^c)/n}} \right), \quad \chi_{ij} = 1 - \epsilon_{ij}, \quad (6)$$

where $C(\gamma) = \log_2(1 + \gamma)$, $V(\gamma) = (1 - (1 + \gamma)^{-2}) (\log_2 e)^2$, $k = B_q$ is the payload in bits, and n is the blocklength. Reliability, payload, and latency are then coupled through $\chi_{ij} = 1 - \epsilon_{ij}$ and $T_{ij} = n/B_c$. The heuristic reliability map is retained only for legacy-result reproduction.

B. DD-Domain Bistatic Sensing Model

For triplet (i, j, q) , define $d_{iq} = \|\mathbf{p}_q - \mathbf{p}_i\|$, $d_{jq} = \|\mathbf{p}_q - \mathbf{p}_j\|$, $\mathbf{u}_{iq} = (\mathbf{p}_q - \mathbf{p}_i)/d_{iq}$, and $\mathbf{u}_{jq} = (\mathbf{p}_q - \mathbf{p}_j)/d_{jq}$. The bistatic delay and Doppler shift are

$$\begin{aligned} \tau_{ijq} &= \frac{d_{iq} + d_{jq}}{c}, \\ \nu_{ijq} &= \frac{(\mathbf{v}_q - \mathbf{v}_i)^T \mathbf{u}_{iq}}{\lambda} + \frac{(\mathbf{v}_q - \mathbf{v}_j)^T \mathbf{u}_{jq}}{\lambda}. \end{aligned} \quad (7)$$

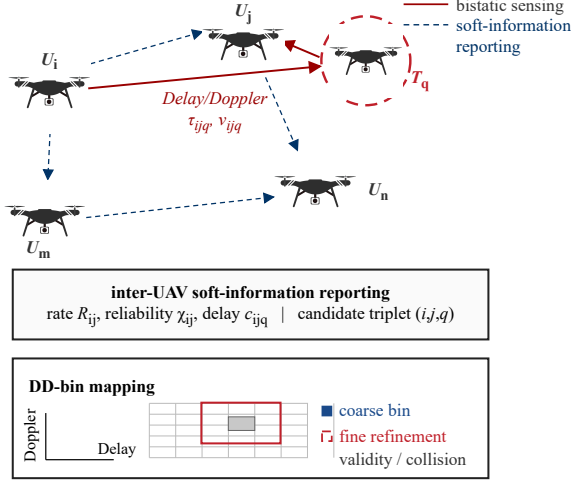
The corresponding DD-grid indices are

$$l_{ijq} = \text{round}(\tau_{ijq} L \Delta f), \quad k_{ijq} = \text{round}(\nu_{ijq} N T). \quad (8)$$

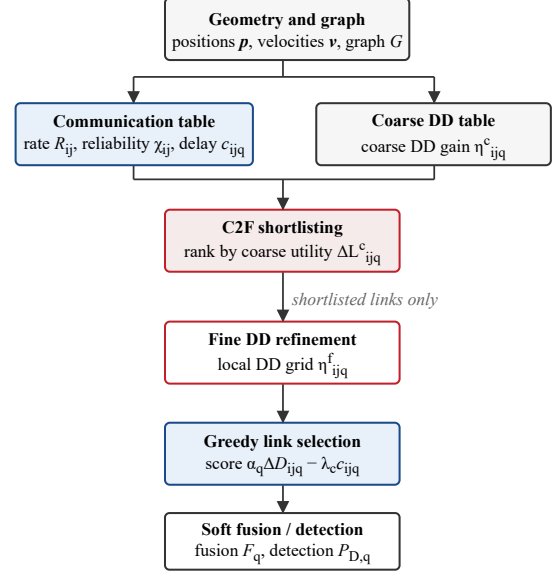
A triplet is DD-valid if $0 \leq l_{ijq} < L$ and $-N/2 \leq k_{ijq} < N/2$. The fractional DD offsets are

$$\varepsilon_{ijq}^T = \tau_{ijq} L \Delta f - l_{ijq}, \quad \varepsilon_{ijq}^V = \nu_{ijq} N T - k_{ijq}. \quad (9)$$

(a) Multi-UAV OTFS-ISAC Scenario



(b) C2F DD-Aware Selection and Fusion



Coarse table: screening; fine table: final selection, fusion, and detection.

Fig. 1. Multi-UAV OTFS-ISAC cooperative sensing and target-specific reporting framework. Each target's reports meet at one fusion UAV selected from the predicted target belief.

Using the normalized sinc function, the coarse DD-domain gain is

$$\eta_{ijq}^c = \text{sinc}^2(\varepsilon_{ijq}^\tau) \text{sinc}^2(\varepsilon_{ijq}^\nu). \quad (10)$$

For C2F DD refinement, a local \$(2W + 1) \times (2W + 1)\$ DD neighborhood provides the waveform-derived recoverable energy \$\eta_{ijq}^{\text{loc}}\$. The refined gain is

$$\eta_{ijq}^f = \eta_{ijq}^{\text{loc}}. \quad (11)$$

Thus, C2F changes computational effort rather than introducing a hand-set sensing-gain interpolation.

If multiple DD-valid targets observed by the same UAV pair fall into the same DD bin, we use the collision penalty

$$\eta_{ijq}^{\text{col}} = \frac{1}{n_{ijq}^{\alpha_{\text{col}}}}, \quad (12)$$

where \$n_{ijq}\$ is the number of targets sharing bin \$(l_{ijq}, k_{ijq})\$.

The bistatic target gain is

$$g_{ijq}^s = \frac{\lambda^2 \sigma_q}{(4\pi)^3 d_{iq}^2 d_{jq}^2}, \quad (13)$$

where \$\sigma_q\$ denotes the target RCS. Combining bistatic gain, processing gain, DD-domain gain, and collision penalty gives

$$\gamma_{ijq}^s = \frac{\tilde{P}_i^s g_{ijq}^s G_p^{\text{DD}} \eta_{ijq}^{\text{col}}}{N_0 + I_j^s}, \quad \eta_{ijq}^{\text{DD}} \in \{\eta_{ijq}^c, \eta_{ijq}^f\}. \quad (14)$$

In the sensing-only power model, \$\tilde{P}_i^s = P_i^s\$ and \$G_p = NL\$. Both receivers use the shared propagation field in (4), with

receiver-specific suppression. The illuminator's direct path remains in \$I_j^s\$, which makes (14) near-far limited rather than interference free.

C. Soft-Information Fusion Model

Each feasible triplet produces a waveform-derived local log-likelihood ratio (LLR). After OTFS matched filtering, the delay-Doppler bin for target \$q\$ on link \$i \rightarrow q \rightarrow j\$ is a single complex sample with a Swerling-I fluctuation. The normalised bin energy \$x = |z|^2 / \sigma_n^2\$ is therefore exponential, \$x \sim \text{Exp}(1)\$ under \$\mathcal{H}_0\$ and \$x \sim \text{Exp}(1 + \gamma_{ijq}^s)\$ under \$\mathcal{H}_1\$. The local log-likelihood ratio, centred under \$\mathcal{H}_0\$, is

$$\ell_{ijq} = \frac{\gamma_{ijq}^s}{1 + \gamma_{ijq}^s} (x - L), \quad (15)$$

where \$L\$ is the number of OTFS frames incoherently integrated in one CPI. Its moments are

$$\begin{aligned} \delta_{ijq} &= \mathbb{E}_1[\ell] - \mathbb{E}_0[\ell] = \frac{L(\gamma_{ijq}^s)^2}{1 + \gamma_{ijq}^s}, \\ \sigma_{ij,0}^2 &= \text{Var}_0[\ell] = \frac{L(\gamma_{ijq}^s)^2}{(1 + \gamma_{ijq}^s)^2}, \end{aligned} \quad (16)$$

The LLR carries no free scale parameter, and its single-link deflection \$\delta_{ijq}^2 / \sigma_{ij,0}^2 = L(\gamma_{ijq}^s)^2\$ is Swerling's classical result.

Communication-error calibration acts on the soft statistic through the reporting leg \$j \rightarrow f_q\$:

$$\delta_{ijq}^{\text{eff}} = \chi_{jfq} \delta_{ijq}, \quad \bar{\sigma}_{ij,0}^2 = \chi_{jfq} \sigma_{ij,0}^2 + (1 - \chi_{jfq}) \sigma_{\text{err}}^2, \quad (17)$$

where $\chi_{j f_q}$ is the success probability (6) of the report $j \rightarrow f_q$. The same post-channel moments generate the selection metric, the detection threshold, and the Monte Carlo samples.

For selected set $\mathcal{S}_q$, the released V1 model assumes conditionally independent local statistics, giving

$$D_q(\mathcal{S}_q) = \sum_{(i,j) \in \mathcal{S}_q} \frac{(\delta_{ijq}^{\text{eff}})^2}{\bar{\sigma}_{ij,0}^2}. \quad (18)$$

Modeling correlated reports is left to future work.

Using the Gaussian deflection approximation, the detection probability is

$$P_{D,q} \approx Q\left(Q^{-1}(P_{\text{FA}}) - \sqrt{D_q(\mathcal{S}_q)}\right), \quad (19)$$

where $Q(\cdot)$ denotes the Gaussian Q -function. Because the local LLR has a positively skewed centred-Gamma null distribution, the Monte Carlo detector calibrates the threshold with the Cornish–Fisher correction $z_{\text{CF}} = z + (\gamma_{1,0}/6)(z^2 - 1)$, where $\gamma_{1,0}$ is computed from the same post-report moments used by fusion.

D. Truth, Belief and the Scheduling Loop

We study prior-assisted confirmation or re-detection after a coarse acquisition stage; blind first acquisition is outside the present scope. The scheduler therefore receives an upstream predicted belief but cannot access the true target state. We carry two states side by side. The *truth* ($\mathbf{p}_{q,t}, \mathbf{v}_{q,t}$) drives echo generation, the true delay-Doppler, the true sensing gain and the detector; the *belief* $b_{q,t} = \mathcal{N}(\hat{\mathbf{x}}_{q,t}, \mathbf{P}_{q,t})$ drives link selection and the DD search window. The intended tracking loop is

$$\begin{aligned} &\text{predict} \rightarrow \text{schedule} \rightarrow \text{sense} \rightarrow \text{update}, \\ &\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1|t-1}, \\ &\mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1|t-1}\mathbf{F}^T + \mathbf{Q}. \end{aligned} \quad (20)$$

The simulator instantiates this belief as an upstream Gaussian snapshot rather than implementing the posterior update. Its covariance is propagated through the bistatic DD Jacobian to form a three-sigma gate. Scheduling uses mean RCS; the current-CPI Swerling fluctuation is marginalized once by (15). Thus, neither target-state nor RCS truth enters scheduling.

III. TARGET-LOCAL ADAPTIVE C2F SELECTION

A. Prediction-Only Fusion Planning

Let $\hat{\mathbf{p}}_q$ be the predicted position supplied by the upstream tracker. V1 assigns target q to

$$f_q = \arg \min_{m \in \mathcal{M}} \|\hat{\mathbf{p}}_q - \mathbf{p}_m^{\text{UAV}}\|_2. \quad (21)$$

The assignment is recomputed from the predicted belief; the true target state and current-CPI RCS realization are unavailable to the scheduler. Unlike the previous target-agnostic rule $\arg \max_m \sum_j R_{jm}$, (21) can map different targets to different fusion UAVs. It therefore changes the reporting quantities $R_{j f_q}$ and $\chi_{j f_q}$ before link selection begins.

For target q , the resulting feasible candidate set is

$$\mathcal{E}_q = \{(i, j) : R_{j f_q} \geq R_{\min}, (l_{ijq}, k_{ijq}) \text{ is DD-valid}\}. \quad (22)$$

The report generated at UAV j is sent to f_q . Under the released serial finite-blocklength MAC, every selected report occupies n/B_c seconds and carries 640 bits.

B. Fair Detector-Aligned Marginal

Let $\mathbf{D} = (D_1, \dots, D_Q)$ collect the communication-calibrated fused deflections. We use the weak-target potential

$$\begin{aligned} U(\mathbf{D}) = & -Q\tau_\alpha \log \sum_{q \in \mathcal{Q}} \exp[-P_{D,q}(D_q)/\tau_\alpha] \\ & - \frac{\mu_\alpha}{2D_{\min}} \sum_{q \in \mathcal{Q}} [D_{\min} - D_q]_+^2. \end{aligned} \quad (23)$$

Its target priority is the gradient

$$\begin{aligned} \alpha_q = & Qw_q \frac{\partial P_{D,q}}{\partial D_q} + \mu_\alpha \frac{[D_{\min} - D_q]_+}{D_{\min}}, \\ w_q = & \frac{e^{-P_{D,q}/\tau_\alpha}}{\sum_r e^{-P_{D,r}/\tau_\alpha}}. \end{aligned} \quad (24)$$

For candidate $e = (i, j, q)$ and current selection $\mathcal{S}$, the implemented greedy replay evaluates

$$\Delta \mathcal{L}_e = U(\mathbf{D}(\mathcal{S} \cup \{e\})) - U(\mathbf{D}(\mathcal{S})) - \lambda_c \tilde{c}_e, \quad (25)$$

where $\tilde{c}_e$ is the reporting latency in milliseconds. The replay uses the same post-channel moments and threshold correction as the Monte-Carlo detector. Starting from $\mathcal{S} = \emptyset$, it commits the largest positive marginal while enforcing $|\mathcal{S}_q| \leq L_{\max}$ and $|\mathcal{S}| \leq L_{\text{tot}}$.

C. Adaptive Two-Stage Refinement

Computing waveform-derived local DD energy for every feasible triplet is unnecessary. The coarse stage evaluates candidates using η_{ijq}^c and maintains a dynamic frontier. Promising candidates are refined with $\eta_{ijq}^f = \eta_{ijq}^{\text{loc}}$; after each commit, affected marginals are refreshed and the fine stage replays the exact detector-aligned rule in (25). This is an adaptive shortlist rather than a fixed one-shot top- K filter. In the V1 main experiment, the procedure evaluates 117.7 local DD neighborhoods per trial, versus 509.3 under full refinement, a 76.9% reduction.

The implemented task objective is

$$F(\mathcal{S}) = U(\mathbf{D}(\mathcal{S})) - \lambda_c \sum_{e \in \mathcal{S}} \tilde{c}_e. \quad (26)$$

The cost term can make F non-monotone, while the soft minimum couples targets and the constraints combine per-target and total caps. We therefore make no universal submodular approximation claim. V1 is an architectural extension of the existing selector: its principal new variable is f_q in (21), not a new greedy score.

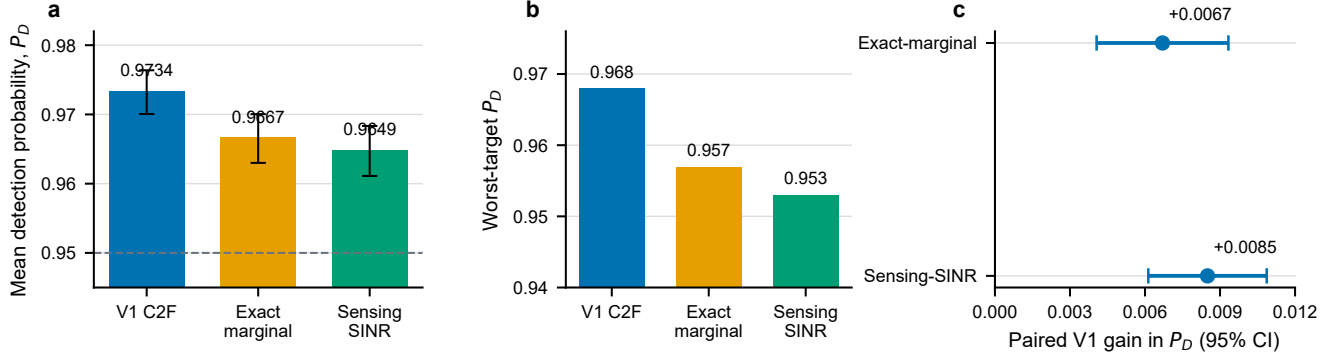


Fig. 2. Confirmatory MC=1000 comparison under identical per-trial resources. (a) Mean detection probability with 95% Wilson intervals; the dashed line marks 0.95. (b) Worst-target detection probability. (c) Paired V1-minus- baseline differences with 95% confidence intervals.

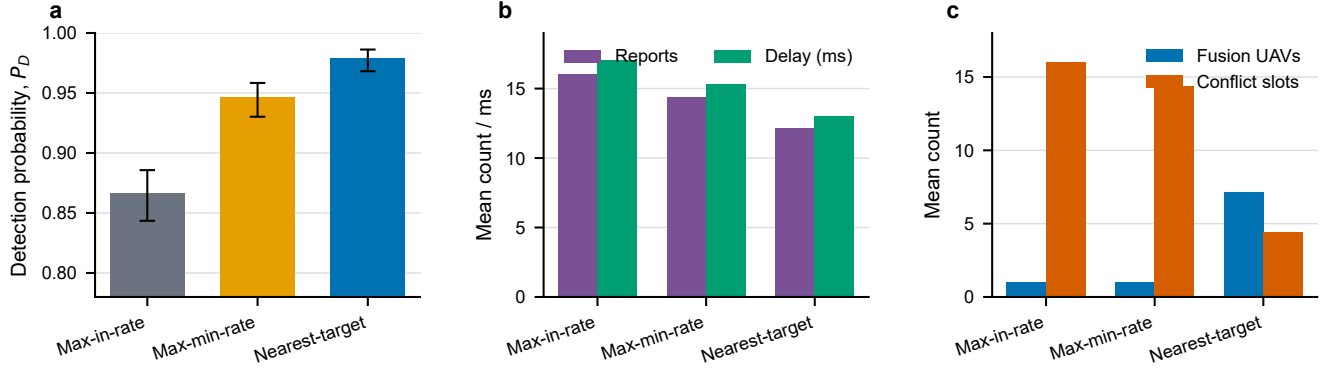


Fig. 3. Fusion-rule mechanism ablation (MC=100 per rule) with the inner adaptive C2F selector fixed. (a) Detection probability and 95% intervals. (b) Selected reports and measured serial delay. (c) Number of assigned fusion UAVs and conflict-graph slots. The slot count diagnoses parallelization potential; it is not an achieved latency.

IV. SIMULATION RESULTS

A. Protocol and Nominal Configuration

We evaluate prior-assisted confirmation in a $4000\text{m} \times 4000\text{m}$ region with $M = 15$ UAVs and $Q = 10$ targets. Table I lists the released V1 configuration. All main methods use the same paired geometries, detector randomness, feasible graph, and per-trial number of selected reports. The confirmatory comparison uses 1000 Monte-Carlo trials. Fusion and operating-envelope sweeps use 100 trials per condition with seed 2026 and are therefore diagnostic rather than substitutes for the main evidence.

B. Confirmatory Main Comparison

Figure 2 compares target-local adaptive C2F with two strong resource-matched rules. Exact-marginal greedy uses the same fair utility but no C2F detector-aligned replay; Sensing-SINR ranks nominal sensing strength. V1 attains $P_D = 0.9734$ (95% CI: [0.9701, 0.9764]), versus 0.9667 for exact-marginal greedy and 0.9649 for Sensing-SINR. Its worst-target detection probability is 0.968, compared with 0.957 and 0.953. The paired gains are +0.0067 (95% CI: [0.0041, 0.0093]) and +0.0085 ([0.0061, 0.0109]), respectively; both interval lower bounds are

TABLE I
TARGET-LOCAL FUSION V1 SIMULATION PARAMETERS

Parameter	Value
UAVs / targets	$M = 15, Q = 10$
Area / communication range	$4000^2 \text{ m}^2, d_{\max} = 2500 \text{ m}$
UAV / target speed	30–60 / 50–90 m/s
Carrier / DD grid	5.9 GHz, $N = L = 64, \Delta f = 30 \text{ kHz}$
Power split / transmit power	$\rho = 0.8, P_i = 1 \text{ W}$
Rate threshold / false alarm	$R_{\min} = 0.2 \text{ Mbit/s}, P_{\text{FA}} = 0.05$
Report / blocklength	640 bits, $n = 2048$ channel uses
Predicted belief	$\sigma_p = 150 \text{ m}, \sigma_v = 15 \text{ m/s}$
Selection	$D_{\min} = 3, \lambda_c = 0.005, L_{\max} = 6, L_{\text{tot}} = 60$
Fusion / reporting	nearest predicted target / serial orthogonal MAC
C2F refinement	$W = 1$, adaptive frontier, detector-aligned replay
Main / sweep trials	1000 / 100 per condition

positive. The empirical active-target and system-wide false-alarm probabilities are 0.0498 and 0.0495, consistent with the designed value 0.05.

Resource matching is exact: each method selects 11.993

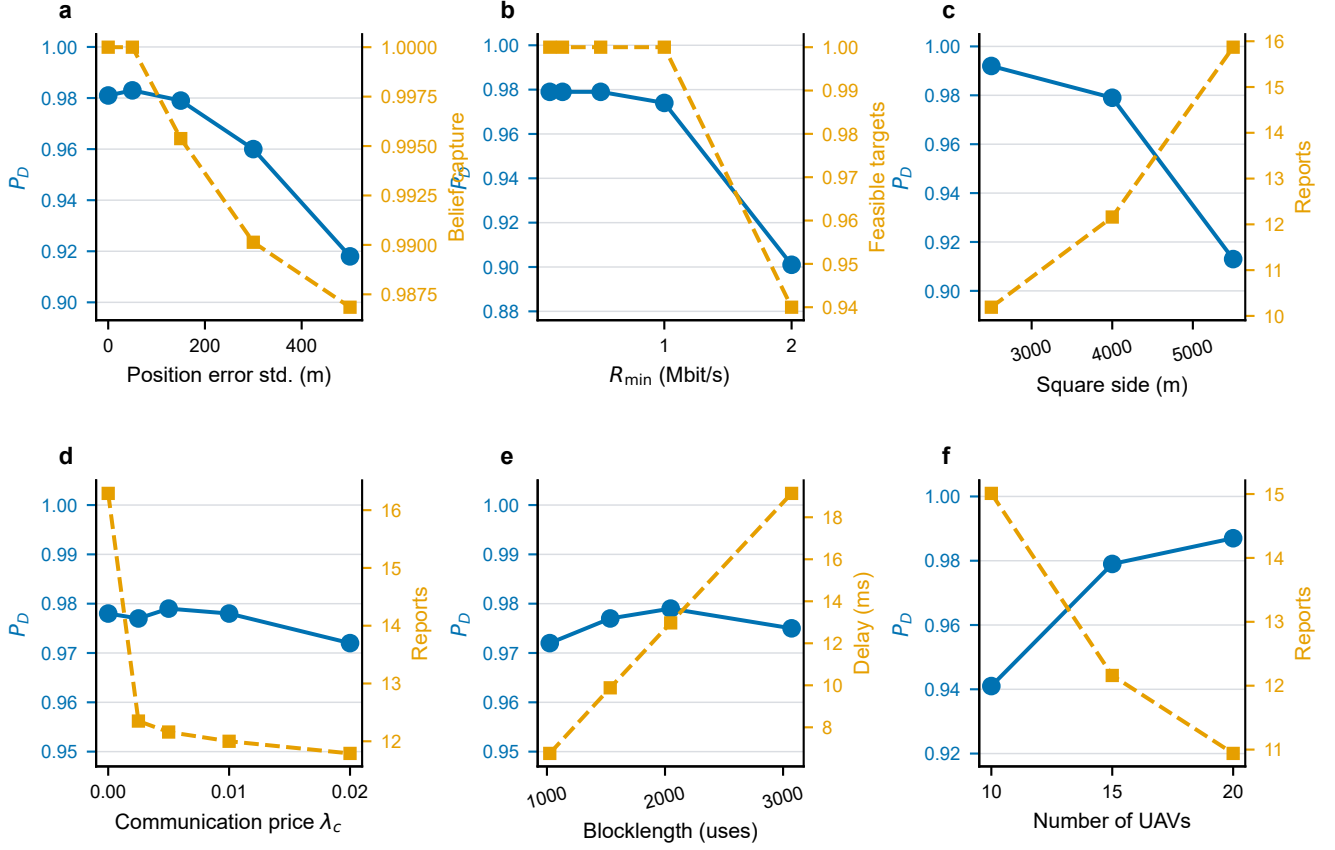


Fig. 4. Exploratory V1 operating envelope (MC=100 per condition). Blue circles show P_D ; orange squares show the named secondary metric. (a) Position-belief error, (b) minimum reporting rate, (c) deployment side length, (d) communication price, (e) blocklength, and (f) UAV fleet size.

reports on average, corresponding to 7.676 kbit and 12.793 ms of serial reporting. Thus, the main detection gain is not purchased with more reports. The adaptive C2F stage evaluates 117.7 fine DD neighborhoods instead of 509.3 for exhaustive local refinement, reducing fine evaluations by 76.9%. These results support the V1 system as a stable nominal point; they do not isolate fusion placement because all three main methods already use the target-local graph.

C. Why Target-Local Fusion Helps

Figure 3 changes only the fusion destination rule. The target-agnostic max-in-rate rule gives $P_D = 0.866$, worst-target $P_D = 0.800$, 15.99 reports, and 17.056 ms. A max-min-rate alternative improves these values to 0.946, 0.930, 14.36, and 15.317 ms. Nearest-target fusion reaches 0.979 and 0.960 with only 12.16 reports and 12.971 ms. Because the two-stage screening, greedy objective, packets, and caps are unchanged, this ablation supports the mechanism in (21): fusion placement changes the reliable reporting-edge set seen by the selector.

The reporting structure explains the difference. Max-in-rate and max-min-rate assign all ten targets to one fusion UAV on average. Nearest-target assigns them across 7.13 UAVs and lowers the mean conflict-graph requirement from 15.99/14.36 to 4.40 slots. If reports in nonconflicting edges could be

sent simultaneously without bandwidth sharing, additional interference, or a changed finite-blocklength model, 4.40 slots would correspond to a 4.693 ms lower bound. V1 does not implement this scheduler, so its reported latency remains the measured serial value 12.971 ms.

D. Operating Envelope and Design Choices

Figure 4 summarizes six controlled scans. First, V1 remains near $P_D = 0.98$ for position-error standard deviations up to 150 m, then decreases to 0.960 at 300 m and 0.918 at 500 m. Belief capture remains 0.9868 even at 500 m; the degradation is therefore more consistent with prediction-induced fusion and link mismatch than with DD-gate misses alone. Second, $R_{\min} \leq 0.5$ Mbit/s is inactive at this nominal geometry. At 2 Mbit/s, only 94% of targets retain feasible candidates and P_D falls to 0.901; the lower report count is candidate starvation, not improved efficiency.

Topology is the next boundary. Increasing the square side from 2500 to 5500 m changes P_D from 0.992 to 0.913 and reports from 10.19 to 15.87. Increasing the fleet from 10 to 20 UAVs at $Q = 10$ changes P_D from 0.941 to 0.987 while reducing reports from 15.01 to 10.94. These scans vary density implicitly and must not be combined into an independent density effect.

The communication price exhibits a broad plateau. Setting $\lambda_c = 0$ requires 16.29 reports at $P_D = 0.978$, whereas $\lambda_c = 0.005$ uses 12.16 reports at $P_D = 0.979$. Values from 0.0025 to 0.01 have overlapping MC=100 uncertainty, so 0.005 is treated as a stable engineering point rather than a statistically unique optimum. Finally, 2048 channel uses yields $P_D = 0.979$ and 12.971 ms. The 1536-use setting is a promising lower-latency candidate (0.977, 9.880 ms), while 3072 uses is dominated in this scan (0.975, 19.136 ms). A paired high-MC comparison is required before promoting the 1536-use setting to a new release.

All figures are generated directly from the released CSV/JSON records. The resolved V1 configuration, source-data index, and plotting command accompany the manuscript, preventing exploratory and confirmatory results from being silently mixed.

V. CONCLUSION

This paper presented target-local fusion V1 for prior-assisted multi-UAV OTFS-ISAC sensing. The outer planner assigns each target to the UAV nearest its predicted position, while the existing communication-calibrated adaptive C2F selector performs detector-aligned greedy replay. The formulation therefore changes where reports meet without changing the inner marginal-gain objective, packet size, blocklength, or resource caps.

In 1000 paired trials, V1 achieved $P_D = 0.9734$ and worst-target $P_D = 0.968$ with 11.993 reports, 7.676 kbit, and 12.793 ms of serial reporting. At identical resources, its paired gains over exact-marginal greedy and Sensing-SINR were 0.0067 and 0.0085, respectively. A controlled fusion-rule ablation showed that target-local planning disperses targets across fusion UAVs and exposes shorter, more reliable reporting edges to the inner selector. The evidence therefore supports a joint architectural conclusion—target-related fusion planning plus adaptive two-stage selection—rather than attributing the full improvement to a new greedy criterion.

V1 is a stable nominal version, not a universal endpoint. Large prediction errors, strict rate thresholds, and sparse deployments remain its principal boundaries. Future work will

validate the main ablations with multiple seeds and 1000-trial paired tests, model obstruction and correlated reports, compare the 1536- and 2048-use operating points, and turn the observed conflict-slot reduction into an interference- and finite-blocklength-aware parallel scheduler.

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